TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2009

BIOLOGY **9747/03**
Higher 2 **Tuesday, 29 September**
Paper 3 **1 hour 30 minutes**

Candidate Name: ____________________________ **Civics Group:** ______ **/ 08**

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions

At the end of the examination, fasten the answer papers securely and hand up separately.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
TOTAL	

This document consists of **13** printed pages.

For Examiner's Use

1 Human growth hormone (hGH) is a peptide hormone containing 191 amino acids, synthesized and released by the cells of the anterior pituitary gland. The hormone stimulates growth and cell reproduction in humans. A deficiency of this hormone in children results in growth failure and short stature. In adults, growth hormone deficiency causes decreased muscle mass and poor bone density.

Fig. 1.1 shows how the gene for human growth hormone (hGH) can be transferred into a bacterium.

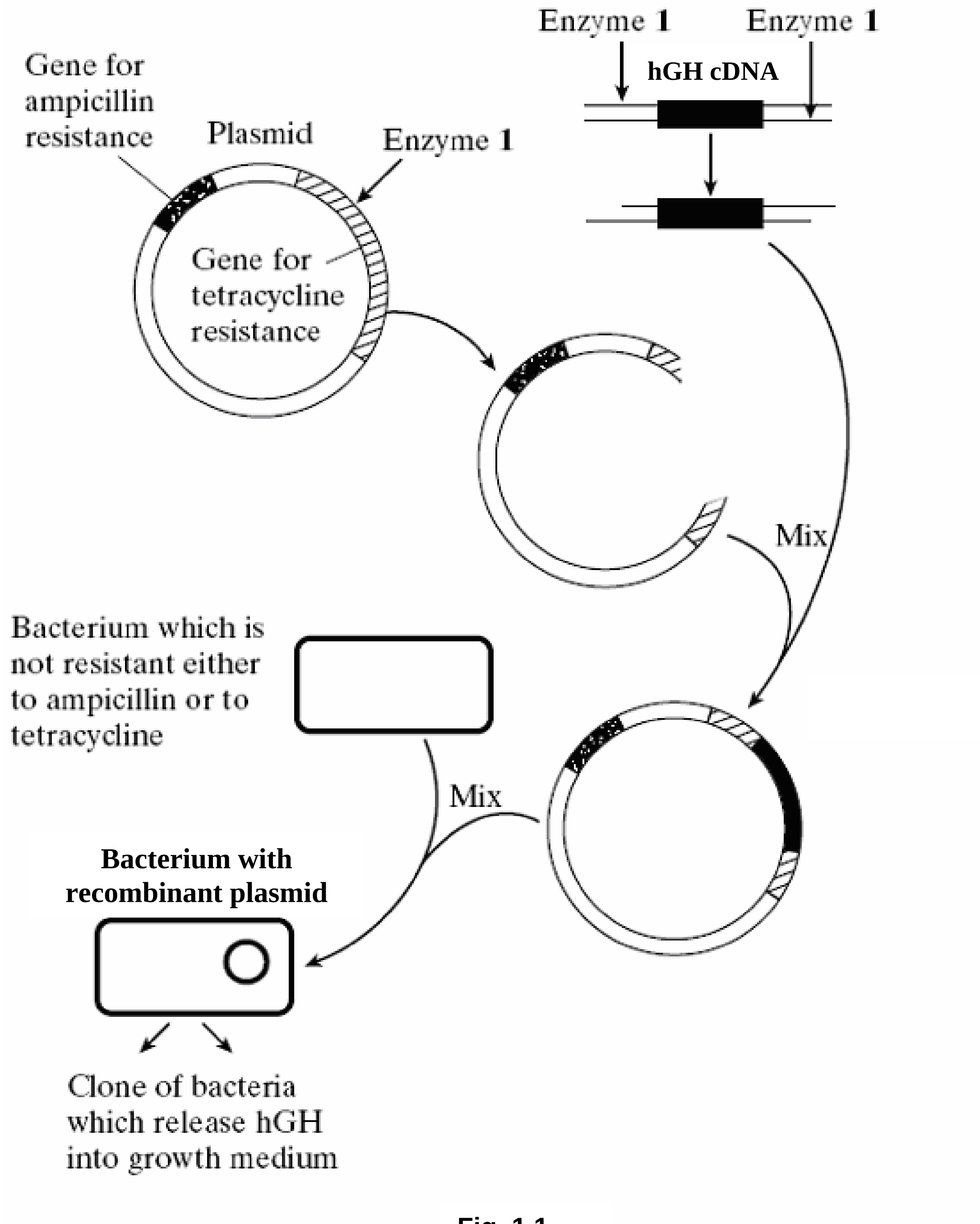

Fig. 1.1

For Examiner's Use

(a) (i) Describe the process by which hGH cDNA is obtained.

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(ii) Explain why **Enzyme 1** is able to cut both the plasmid DNA and cDNA molecules.

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(iii) Describe the steps involved in introducing the recombinant plasmid into the bacterium.

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(iv) Explain **one** possible danger of using plasmids which contain antibiotic resistance genes in genetic engineering.

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[Turn over

Phage viruses are commonly used as a means to transfer foreign DNA into bacteria cells. When a phage infects a layer or lawn of bacterial cells that are growing on a flat surface, a zone of lysis or growth inhibition can occur. This produces a clear zone in the bacterial lawn, known as a plaque. Each plaque originates from a single phage particle.

Fig. 1.2 illustrates the process involved in bacterial transformation and selection using phage vectors.

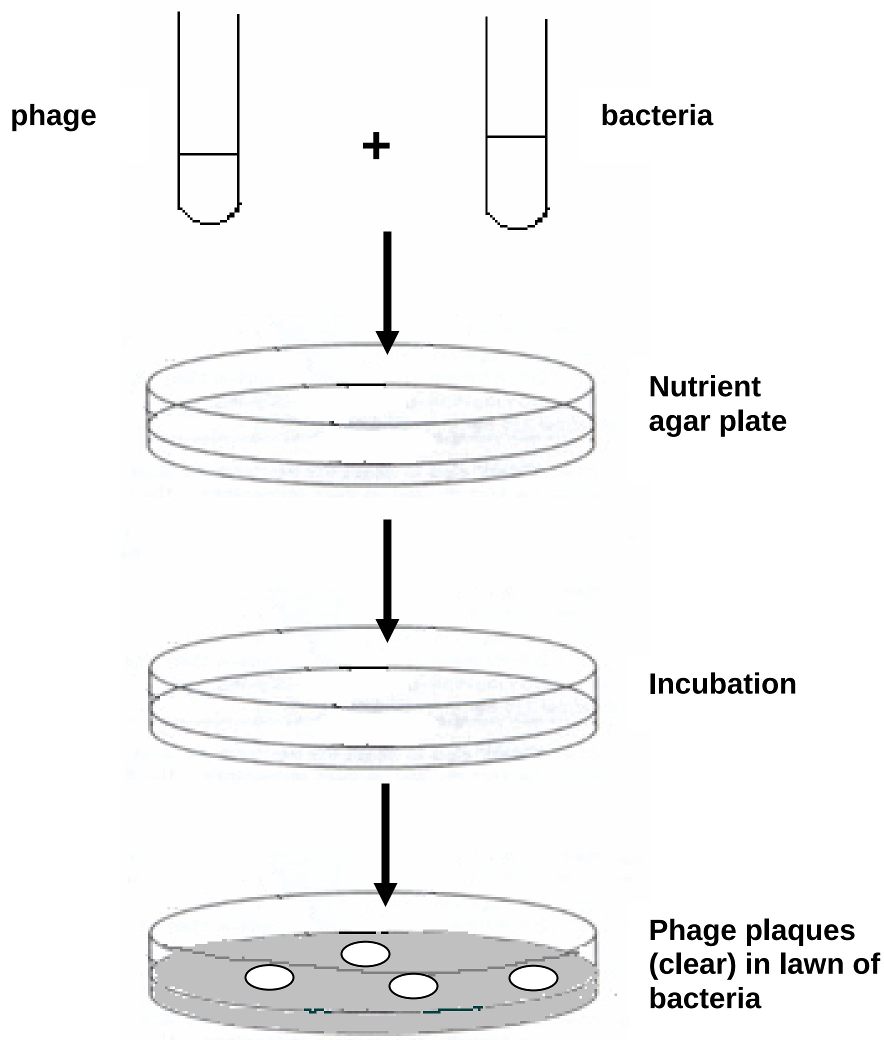

Fig. 1.2

For Examiner's Use

(b) **(i)** With reference to a specific example, describe how a virulent phage can be used as a vector for gene cloning in bacteria.

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(ii) Suggest **two** advantages of the use of phage vectors over plasmid vectors in transferring DNA into bacteria cells.

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[Total: 15]

For Examiner's Use

2 Plants have been used for centuries as medicines - now genetically modified plants can produce plastic, skin tissue agents and human blood proteins. Biotechnologists have created genetically modified (GM) plants that can grow plastic. Conventional plastics are made from oil and do not degrade easily, but the plant plastic is biodegradable. To create these plants, four genes from plastic-producing bacteria were inserted into varieties of oilseed rape (Eurasian plant) and cress (as shown in Fig. 2.1). Meristematic cells (i.e. undifferentiated cells found in zones where cell division and growth can take place) were targeted for the insertion of the desired genes after many failed attempts using various other cell types.

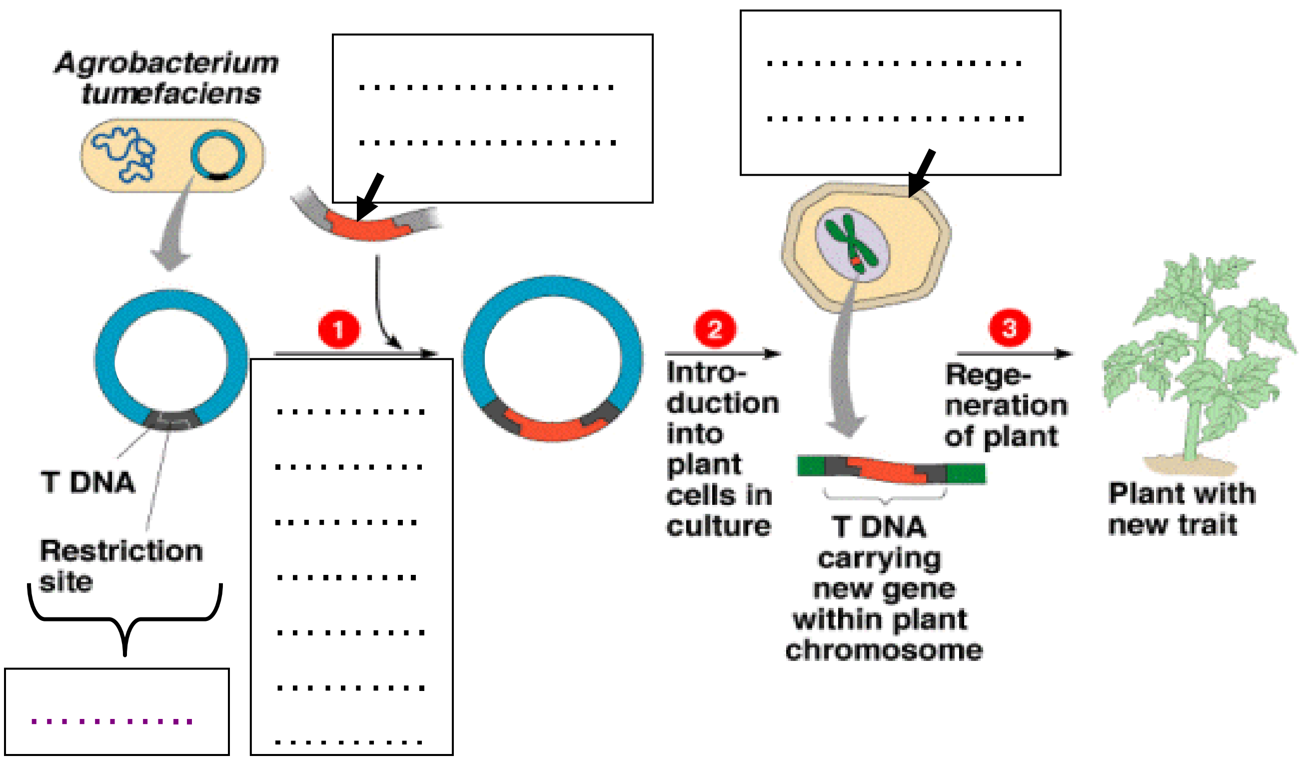

Fig. 2.1

(a) Outline the procedure for creating the plastic-producing plants by completing the boxes provided in Fig. 2.1.

[4]

(b) Selection for transformed plant cells was achieved through observation of a blue coloration in the cell masses. With reference to Fig. 2.1, state when and explain how selection was carried out.

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(c) Explain why using meristematic cells resulted in a higher success rate.

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Golden Rice, a variety of *Oryza sativa* rice, produced through genetic engineering to biosynthesize beta-carotene, was developed as a fortified food to be used in areas where there is a shortage of dietary vitamin A. The *psy* (from daffodils) and *crt1* (from a soil bacterium) genes were transformed into the rice nuclear genome and placed under the control of an endosperm specific promoter, so that they are only expressed in the endosperm. Upon successful transformation, the plant cells are cultured as shown in Fig. 2.2.

In 2005 a new variety called *Golden Rice 2* was announced which produces up to 23 times more beta-carotene than the original variety of golden rice. Neither variety is currently available for human consumption.

Fig. 2.2

(d) State **one** advantage and **one** disadvantage of using the technique shown in Fig. 2.2 for the production of the Golden Rice.

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……[2]

(e) Explain why "neither variety is currently available for human consumption".

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Initial research during the production for Golden Rice involved using *psy* and *crt1* genes obtained through polymerase chain reaction. However, these fragments had blunt ends and thus the concentration of DNA ligase had to be increased to achieve more recombinant plasmids.

(f) Suggest **one** possible advantage of using blunt end fragments for cloning.

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……[2]

(g) Besides increasing the concentration of DNA ligase, suggest **one** other way to overcome the problem of using blunt end fragments in cloning.

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……[2]

[Total: 15]

For Examiner's Use

3 Restriction fragment length polymorphisms (RFLPs) have been used in genomic mapping since the 1980s.

(a) Define restriction fragment length polymorphism.

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(b) Suggest **two** ways in which RFLPs might be useful in genomic mapping.

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(c) In a study of gene **X** in mice, some scientists decided to use two well-studied RFLP loci, RFLP locus **A** and RFLP locus **B**, to determine the relative position of gene **X** on chromosome 1. Fig. 3.1a shows the relative positions of the two RFLP loci on the chromosome 1, and Fig. 3.1b shows the alleles present at each RFLP locus.

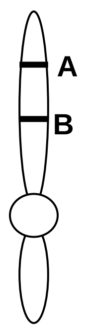

Fig. 3.1a

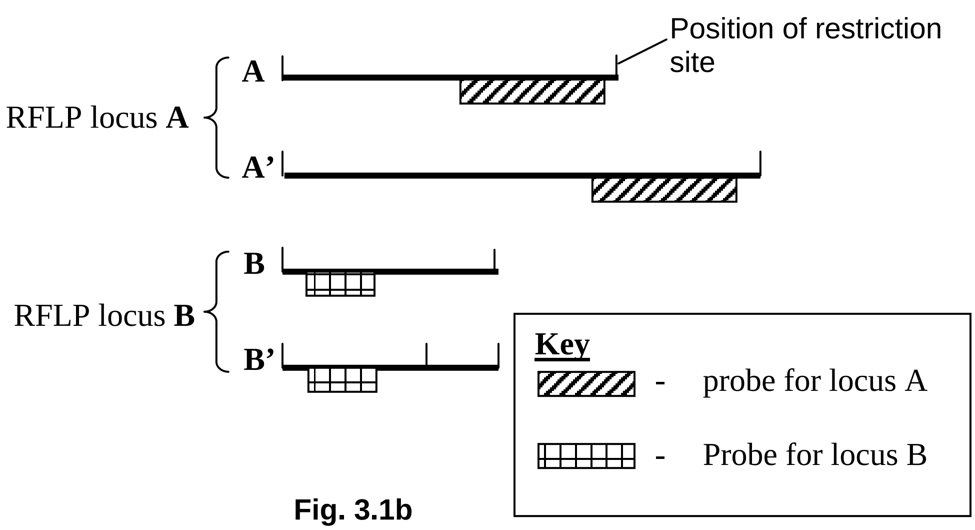

Fig. 3.1b

(i) With reference to Fig. 3.1b, suggest how RFLPs might arise.

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To determine the recombination frequency between RFLP loci **A** and **B**, the researchers carried out a few crosses. In cross **1**, mice homozygous for alleles **A** and **B** were crossed with mice homozygous for alleles **A'** and **B'**. Fig. 3.2 shows the autoradiograph from mice homozygous for alleles **A** and **B**.

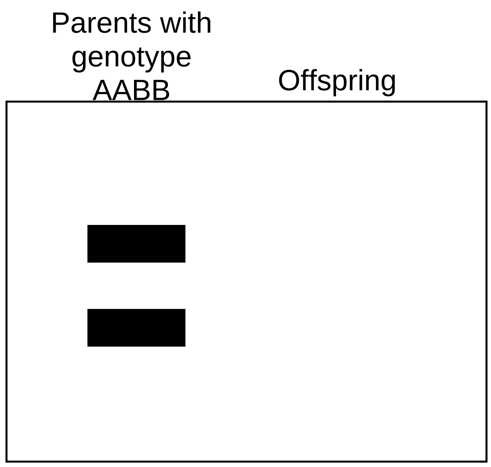

Fig. 3.2

(ii) Explain the term recombination frequency.

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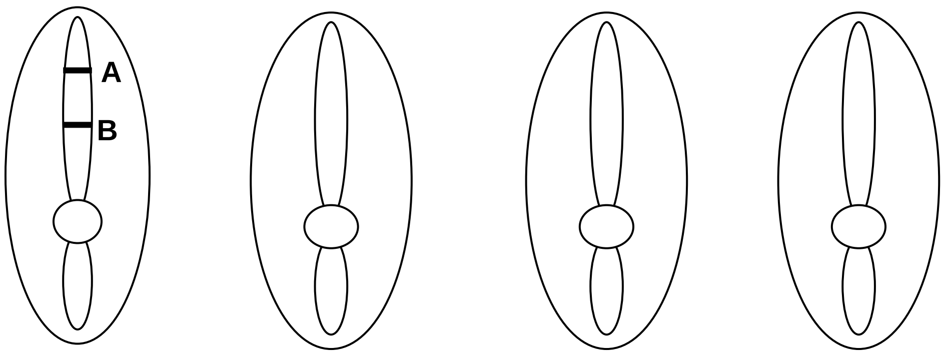
(iii) On Fig. 3.2, draw the positions of the bands you would expect to see upon analysing the DNA of the offspring from cross **1**.

[1]

(iv) Fig. 3.3 shows one possible type of gamete that the offspring from cross **1** can form. Complete the diagram by showing all other possible types of gametes that can be formed by the offspring from cross **1**.



[1]

Fig. 3.3

In cross **2**, the offspring from cross **1** were crossed with mice homozygous for alleles **A** and **B**. RFLP analysis of the offspring from cross **2** was carried out, and four distinct RFLP profiles were obtained (shown in Fig. 3.4).

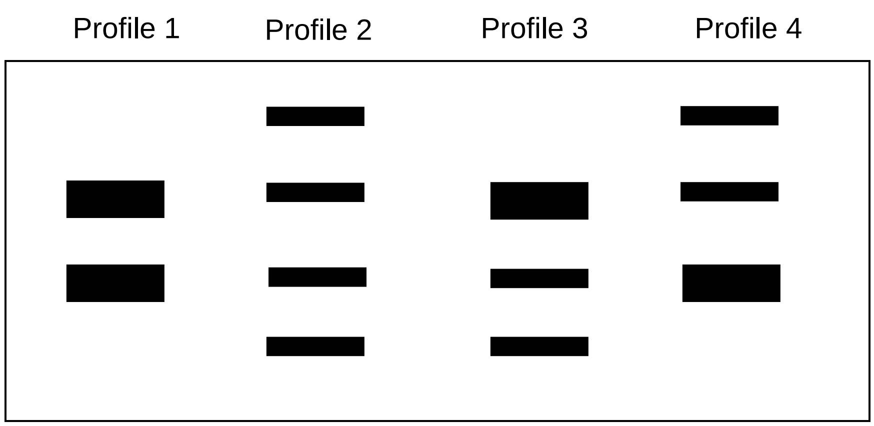


Fig. 3.4

The number of mice with each RFLP profile is as follows:

Profile	No. of mice
1	464
2	471
3	103
4	97

(v) Calculate the recombination frequency of RFLP loci **A** and **B**, using the given formula, and show your working in the space provided.

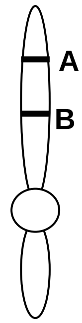
$$\text{Recombination frequency} = \frac{\text{total no. of recombinant offspring}}{\text{total no. of offspring}}$$

[2]

In two separate sets of crosses, researchers worked out the recombination frequencies between gene **X** and RFLP locus **A**, as well as that of gene **X** and RFLP locus **B**. These recombination frequencies are given below:

Recombination frequency between **A** and **X**: 3.4%
Recombination frequency between **B** and **X**: 14.2%

(vi) In the diagram given below, indicate the relative position of gene **X**.



[1]

(d) State **two** other applications of RFLP analysis.

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[Total: 15]

Write your answers to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4 **(a)** Relate the features of adult stem cells to their use in the treatment of leukemia. [8]

(b) Using specific example(s), compare viral and non-viral delivery systems in the treatment of cystic fibrosis. [6]

(c) Describe the process of Restriction Fragment Length Polymorphism (RFLP) detection. [6]